

2025 AMC 10B Problem 2 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 2. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 2

Problem:

Jerry wrote down the ones digit of each of the first 2025 positive squares: 1, 4, 9, 6, 5, 6, . . . What is the sum of all the numbers Jerry wrote down?

Answer Choices:

- A. 9025
- B. 9070
- C. 9090
- D. 9115
- E. 9160

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Solution:

Jerry wrote down the ones digit of each of the first 2025 positive squares:

$$1^2, 2^2, 3^2, \dots, 2025^2.$$

The ones digit of n^2 depends only on the ones digit of n . For $n = 0, 1, \dots, 9$ we have

ones digit of n	0	1	2	3	4	5	6	7	8	9
ones digit of n^2	0	1	4	9	6	5	6	9	4	1

So in each block of 10 consecutive integers, the ones digits of the squares are

$$1, 4, 9, 6, 5, 6, 9, 4, 1, 0,$$

whose sum is

$$1 + 4 + 9 + 6 + 5 + 6 + 9 + 4 + 1 + 0 = 45.$$

We are considering the first 2025 squares, i.e. 1^2 through 2025^2 . Since

$$2025 = 202 \cdot 10 + 5,$$

there are 202 full blocks of 10 squares, plus 5 extra squares. The 202 full blocks contribute

$$202 \cdot 45 = 9090$$

to the sum of the ones digits. The remaining 5 squares are 2021^2 , 2022^2 , 2023^2 , 2024^2 , 2025^2 , which have the same ones digits as 1, 4, 9, 6, 5. Their sum is

$$1 + 4 + 9 + 6 + 5 = 25.$$

Therefore, the total sum of the ones digits Jerry wrote down is

$$9090 + 25 = \boxed{(D) \ 9115}.$$

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